

Advanced Water Safety

Data Analysis Activity Pack

Three self-contained data analysis activities using real Victorian and global drowning statistics. Each activity can be used as a lesson starter, homework task, or standalone assessment item.

Year 9–10 HPE

3 standalone activities

Real data

VC2HP10P02 • VC2HP10P08

~15 min each

About this pack: These three activities use real, current Victorian and global drowning data. Each activity is self-contained and includes a worked example to get you started. Activities can be used in any order. Verify current data at lsv.com.au/research/victorian-drowning-reports/ and royallifesaving.com.au/research.

ACTIVITY A

Trend Analysis — Victorian Drowning Rates Over 5 Years

~15 min

VC2HP10P08

Individual

Source: lsv.com.au

The table below shows Victorian drowning death totals from 2019–20 to 2023–24. Use this data to calculate rates, identify trends, and interpret the implications.

Financial Year	Fatal Drownings (Victoria)	Vic. Population (approx.)	Rate per 100,000 (calculate)	Above/Below 5-yr average?
2019–20	46	6.63M	-----	-----
2020–21	42	6.65M	-----	-----
2021–22	47	6.68M	-----	-----
2022–23	50	6.80M	-----	-----
2023–24	54	6.90M	-----	-----
5-year average	-----	—	-----	—

Worked example (2019–20):

Rate = fatal drownings ÷ (population ÷ 100,000)
= 46 ÷ (6,630,000 ÷ 100,000) = 46 ÷ 66.3 = **0.69 per 100,000**

Tip: divide population by 100,000 first, then divide deaths by that number.

Question A.1 — Trend Description

2 marks

Describe the trend in Victorian drowning deaths from 2019–20 to 2023–24. Is the trend increasing, decreasing, or fluctuating? Use data from the table to support your description.

Question A.2 — Trend Interpretation

3 marks

Identify ONE factor that might explain why Victorian drowning deaths increased from 2020–21 to 2023–24. Your explanation should refer to a specific social, environmental, or policy factor — not just "more people went swimming."

Source or evidence to support your explanation: _____

Question A.3 — Graph Task

3 marks

In the space below, construct a line graph showing Victorian drowning rates per 100,000 from 2019–20 to 2023–24. Label both axes, include a descriptive title, and add a horizontal line showing the 5-year average rate.

✎ Draw your line graph here — or attach on a separate page. Include: title, labelled axes, data points connected by a line, and a dashed horizontal line for the average.

Graph title: _____

2-sentence interpretation: _____

Question A.4 — Prediction

2 marks

Based on the trend you identified, predict Victoria's approximate drowning rate in 2025–26 if no new prevention strategies are implemented. Justify your prediction with reference to the trend data.

My prediction: approximately _____ per 100,000 (_____ deaths)

ACTIVITY B

CALD Over-Representation — Structured Argument Analysis

⌚ ~15 min

VC2HP10P02

Individual / Pairs

Source: lsv.com.au

Use the data below to analyse CALD community over-representation in Victorian drowning deaths. You will calculate the over-representation ratio, interpret it, and construct a structured argument.

Group	% of Victorian Drowning Deaths 2023–24	% of Victorian Population (approx.)	Ratio (deaths % ÷ population %)
CALD community members	~40%	~28%	_____
Non-CALD Victorians	~60%	~72%	_____

Worked example — Over-representation ratio:

Ratio = deaths % ÷ population % = $40 \div 28 = 1.43$

Interpretation: CALD community members are drowning at 1.43 times the rate you would expect based on their population share. A ratio of 1.0 would mean exactly proportional representation.

Question B.1 — Calculate & Interpret

2 marks

Calculate the over-representation ratio for non-CALD Victorians. What does a ratio below 1.0 tell you about non-CALD drowning risk relative to their population share?

Non-CALD ratio: $60 \div 72 =$ _____

A ratio below 1.0 means: _____

Question B.2 — Structural vs Individual Argument

4 marks

Construct TWO arguments explaining the CALD over-representation: one individual-level and one structural-level. Then explain which is more useful for developing prevention policy, and why.

INDIVIDUAL-LEVEL ARGUMENT (BLAMES PERSONAL CHOICES)

Claim:

Because:

Therefore:

STRUCTURAL-LEVEL ARGUMENT (IDENTIFIES SYSTEM FACTORS)

Claim:

Because:

Therefore:

The structural argument is more useful for prevention policy because: _____

Question B.3 — Prevention Proposal

3 marks

Propose ONE specific, evidence-based prevention strategy that directly addresses a structural barrier for CALD community members. Your proposal must include: what it is, who delivers it, how it addresses a structural barrier, and a measurable success metric.

MY PREVENTION PROPOSAL

Strategy:

Who delivers it:

Structural barrier addressed:

Success metric:

ACTIVITY C

Global Comparison — Australia vs Two Other Countries

⌚ ~15 min

VC2HP10P09

Individual

Source: who.int · cdc.gov · royallifesaving.com.au

Use the data below and your own research to complete an international comparison of drowning rates and prevention approaches. Complete the data table, then answer the analysis questions.

Country	Approx. drowning rate (per 100,000)	Income category (World Bank)	Key prevention approach	Source URL
 Australia	1.2	High income		
 Global average (WHO)	~4.0	Mixed		who.int
Country of your choice 1: Research and fill in				
Country of your choice 2: Research and fill in				

Question C.1 — Income and Drowning Risk

3 marks

The WHO reports that approximately 90% of global drowning deaths occur in low- and middle-income countries. Using your data table and the WHO global average, explain the relationship between income level and drowning risk. Why is this relationship NOT simply explained by "low-income people take more risks"?

Source(s): _____

Question C.2 — What Can Australia Learn?

3 marks

Look at the prevention approaches of the two countries you researched. Identify ONE element from either country that Australia does NOT currently emphasise, and evaluate whether Australia should adopt it. Use the structured argument framework.

INTERNATIONAL ELEMENT TO CONSIDER

Element:

From (country):

Why it works there:

Should Australia adopt it? (Yes/No/Modified):

Justification:

Question C.3 — Reflection

2 marks

What does the international comparison tell you about the effectiveness of Australia's current water safety approach? Is Australia performing well relative to comparable high-income countries, or is there room for significant improvement?

The following data relates to diving-related aquatic spinal injury in Australia. Unlike drowning data (which is reported annually), spinal injury data is compiled across multi-year studies. Use this data alongside the Victorian Drowning Report data in Sections 1–3.

10%

of all spinal cord injuries in Australia caused by unsafe diving

Source: AIHW

20%

of quadriplegia cases in Australia result from diving injuries

Source: AIHW

95%

of aquatic spinal injury patients are male

Source: Royal Perth Hospital spinal injury review

75%+

of aquatic spinal injury patients are under 30 years of age

Source: Royal Perth Hospital spinal injury review

66%

of aquatic spinal injury patients suffer permanent disability

Source: Royal Perth Hospital spinal injury review

10–14

years: highest incidence age group for diving-related spinal injuries

Source: AIHW

Post-COVID Youth Drowning Trend

Period	Age group	Drowning rate	Change
5 years pre-COVID	15–20 year olds	Baseline	—
5 years post-COVID	15–20 year olds	+34% above baseline	↑ 34%

Source: Royal Life Saving Society Australia (RLSSA) research. [aihw.gov.au](#) · [royallifesaving.com.au](#)

Prevention context: The core Australian response to this data is the "**Feet First, First Time**" message — always enter unknown water feet-first, and check depth on every visit regardless of prior experience. Inland waterway locations (rivers, dams, lakes) carry the highest drowning risk of any environment in Australia and present additional hazards beyond diving injury: cold water shock, poor visibility, and remote rescue delay.

Q4.1 — Data Comparison

3 marks

The Victorian drowning rate for 2023–24 was 0.78 per 100,000 population. Diving spinal injuries affect a significantly different population profile. Compare the demographic distribution of Victorian drowning victims with the demographic distribution of aquatic spinal injury patients. What does this comparison reveal about the limitations of using drowning data alone to understand the full scope of dangerous water behaviour among young Australians?

Q4.2 — Rate Calculation Challenge

3 marks

The data shows 95% of aquatic spinal injury patients are male. If we assume the overall aquatic spinal injury population is evenly distributed

across the age 10–30 male population in Australia (approximately 3.2 million), and that there are approximately 100 new cases per year nationally:

(a) Calculate the rate per 100,000 for males aged 10–30. (b) How does this compare to the Victorian drowning rate (0.78 per 100,000 all ages)?

(c) What are the significant limitations of this calculation?

Q4.3 — Causal Mechanism

4 marks

The post-COVID data shows a 34% increase in drowning among 15–20 year olds. Propose and evaluate THREE possible explanations for this increase. For each explanation, identify: (a) what evidence would support it, and (b) what prevention response would follow if it were the primary cause.